Optimization in Deep Learning

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ICT, CAS

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Note:

1. □□□□

Optimization Problem & Deep Learning

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Optimization Problem Definition

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$$
\begin{aligned}
\min_{x \in \mathbb{R}^n} \quad & f(x) \\
\text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m \\
& h_i(x) = 0, \quad i = 1, \ldots, p
\end{aligned}
$$

□□ $f(x)$ □□□□□□□x □□□□□□□s.t. □□□□□□□□□

Optimization Problem & Deep Learning

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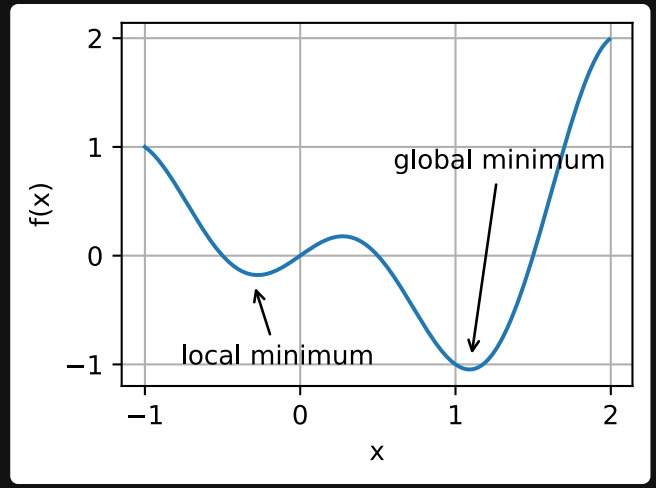
Challenges in Deep Learning Optimization

1. ⊠⊠⊠⊠⊠(local minimum)

⊠⊠⊠⊠⊠⊠⊠⊠ $f(x)$⊠⊠⊠⊠ x ⊠⊠⊠⊠⊠ $f(x)$ ⊠⊠⊠ x ⊠⊠⊠⊠⊠⊠⊠⊠ $f(x)$ ⊠⊠⊠⊠ $f(x)$ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ $f(x)$ ⊠⊠ x ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ $f(x)$ ⊠⊠⊠⊠⊠⊠



Challenges in Deep Learning Optimization

2. ☒☒(saddle point)

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Challenges in Deep Learning Optimization

3. ⊠⊠⊠⊠

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Challenges in Deep Learning Optimization

[illegible]

Basic Definition

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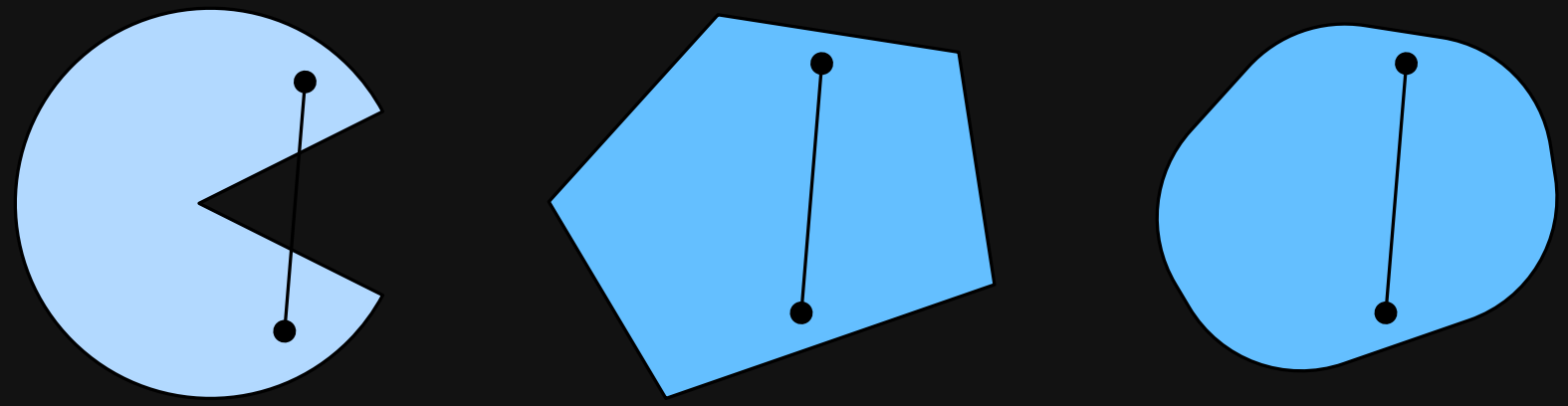
Convex Set

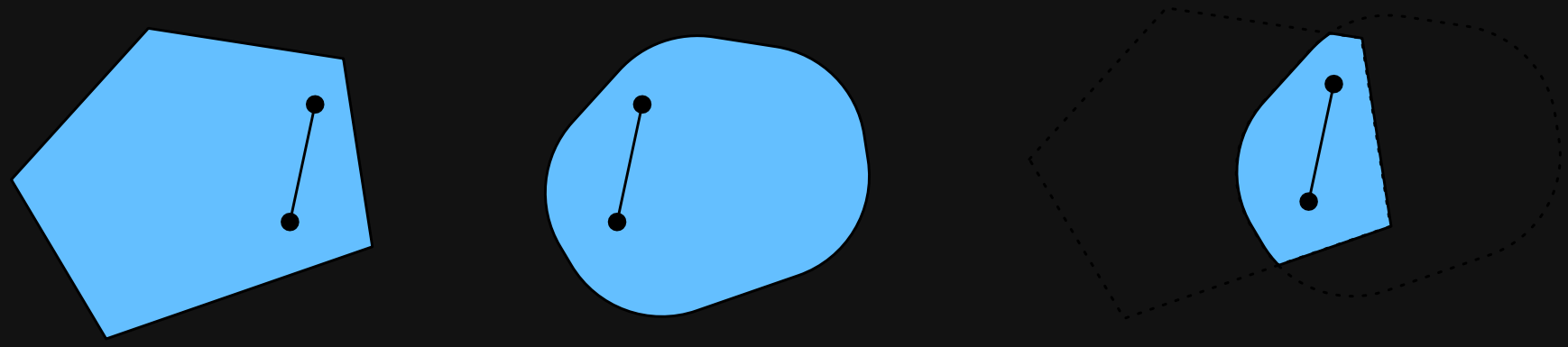
□□□□ $\mathcal{X}$ □□□□□□□□□□□□□ $x, y \in \mathcal{X}$ □ $\alpha \in [0, 1]$□□□□

$$\alpha x + (1 - \alpha) y \in \mathcal{X}$$



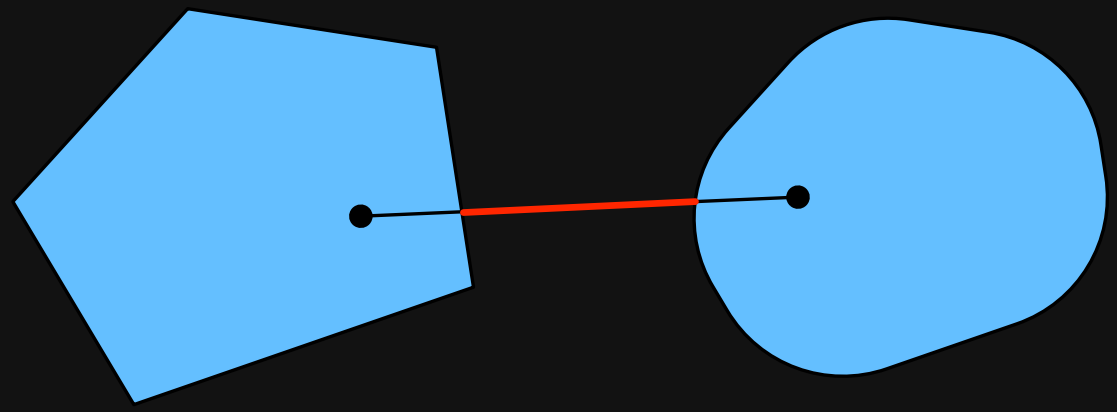
Convex Set

□□□□□□ $\mathcal{X}$ □ $\mathcal{Y}$□□□□□□□□□ $\mathcal{X} \cap \mathcal{Y}$ □□□□□



Convex Set

□□□□□□□□□□□ $\mathcal{X}$ □ $\mathcal{Y}$□□□□□□□□□ $\mathcal{X} \cup \mathcal{Y}$ □□□□□□□

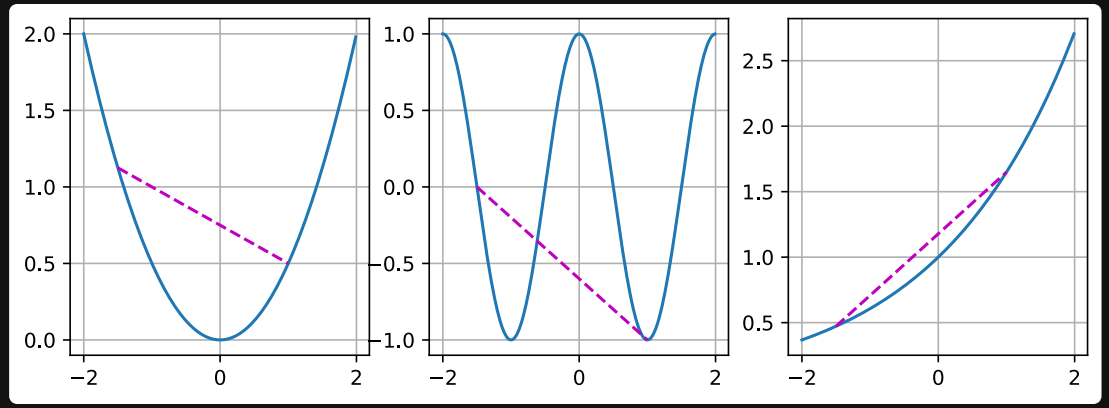


Convex Function

□□□□□□ $\mathcal{X}$□□□□□□□□ $x, y \in \mathcal{X}$ □ $\alpha \in [0, 1]$□□□□

$$f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha) f(y)$$

□□□ f □□□□□



Properties of Convex Function

1. ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

⊠⊠⊠⊠ f ⊠⊠⊠⊠⊠⊠⊠ $x^* \in \mathcal{X}$ ⊠ f ⊠⊠⊠⊠⊠⊠⊠⊠ x^* ⊠⊠ f ⊠⊠⊠⊠⊠⊠

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⊠⊠ x^* ⊠ f ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ p⊠⊠⊠⊠ $x \in \mathcal{X}$ ⊠ $0 < |x - x^*| < p$ ⊠⊠⊠⊠ $f(x) \geq f(x^*)$□

⊠⊠⊠⊠ x^* ⊠⊠ f ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ $x \in \mathcal{X}$ ⊠⊠ $f(x) < f(x^*)$⊠⊠ $\alpha = 1 - \frac{p}{|x - x^*|}$ ⊠⊠⊠⊠⊠⊠

$$
\begin{aligned}
f(\alpha x + (1-\alpha)x^*) &\leq \alpha f(x) + (1-\alpha) f(x^*) \\
&< \alpha f(x^*) + (1-\alpha) f(x^*) \\
&= f(x^*)
\end{aligned}
$$

⊠⊠ x^* ⊠ f ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ x^* ⊠ f ⊠⊠⊠⊠⊠⊠

Properties of Convex Function

2. ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

3. ⊠⊠⊠⊠⊠⊠⊠⊠

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Strong Convex

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Strong Convex

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Lipschitz Continuity

L-Lipschitz ⊠⊠⊠

Smoothness

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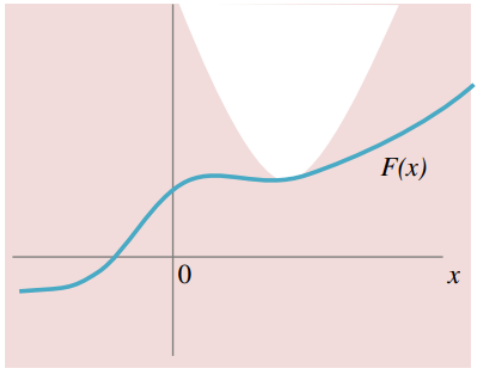
⊠⊠⊠⊠⊠⊠ $f : R^d \to R$⊠⊠⊠ $||\cdot||$⊠⊠⊠⊠⊠⊠⊠ $L > 0$⊠⊠⊠⊠⊠⊠⊠ $x, y \in R^d$□□□□□□□□□□□□

$$f(x) - f(y) \leq \nabla f(y)^\top (x - y) + \frac{L}{2}||x - y||^2$$

⊠⊠⊠⊠ f ⊠⊠⊠ $||\cdot||$ ⊠ L-⊠⊠⊠⊠

Fig. 2.2 Illustration of Lipschitz smoothness. If a scalar function $F(x)$ is L-Lipschitz smooth, then for any point x, the function lies inside the shaded region shown in the picture, as specified by (2.2)



Smoothness

□□□□□□□

$$|\nabla f(x) - \nabla f(y)| \leq L||x - y||$$

□□□□□f□L-□□□□□□□□□□□□□∇f□L-Lipschitz □□□□

Gradient Decent & Mini-batch Stochastic Gradient Decent

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Gradient Decent

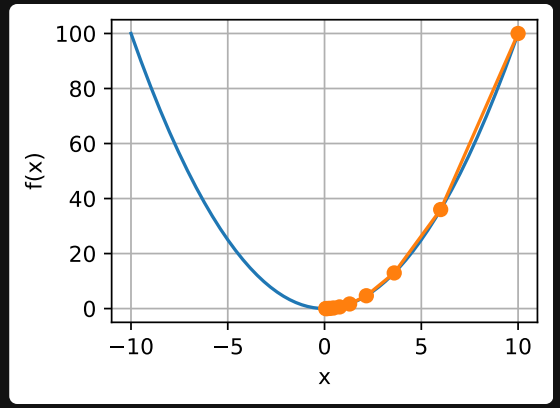
□□□□□□□□□

$$w_{t+1} = w_t - \eta_t \nabla f(w_t)$$

□□□η_t ☒☒☒☒☒w_t ☒☒☒☒$f(w_t)$ ☒☒☒☒☒☒☒ $\eta_t \leq \eta_{t-1} \leq \cdots \leq \eta_1$ ☒☒☒☒☒

□□□□□□□□□□□□□□□

$$\begin{aligned} \|w_{t+1} - w^*\|^2 &= \|w_t - \eta_t \nabla f(w_t) - w^*\|^2 && (1) \\ &= \|w_t - w^*\|^2 - 2\eta_t \langle \nabla f(w_t), w_t - w^* \rangle + \eta_t^2 \|\nabla f(w_t)\|^2 && (2) \end{aligned}$$



Gradient Decent

□□□□□□□□□□□□□□□□

□□ $f(x)$ □ x □□□□□□□□□□□□□□

$$f(x+\epsilon) = f(x) + \epsilon f'(x) + o(\epsilon^2)$$

□□□□□□□□ $\eta > 0$□□ $\epsilon = -\eta f'(x)$□□□□

$$f(x - \eta f'(x)) = f(x) - \eta f'(x)^2 + o(\eta^2 f'(x)^2)$$

□□ $f'(x) \neq 0$□□□□□□□□□□ η □□□□□□ $o(\eta^2 f'(x)^2)$ □□□□□□□□□

$$f(x - \eta f'(x)) < f(x)$$

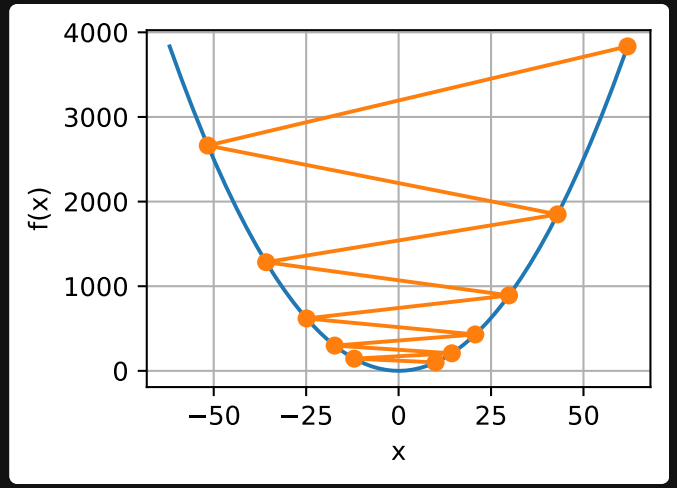
Gradient Decent (Learning Rate)

□□□□□□□□

1. □□□□□□□□□

Gradient Decent (Learning Rate)

2. □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□



Gradient Decent (Learning Rate)

3. □□□□

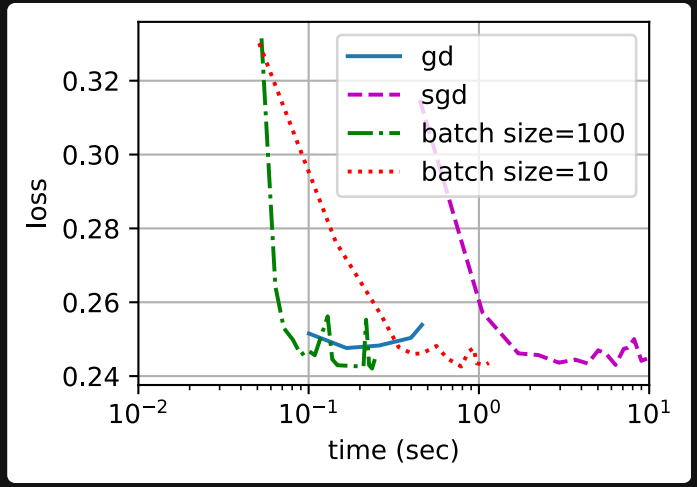
Stochastic Gradient Decent

□□□

Mini-batch Stochastic Gradient Decent

Mini-batch Stochastic Gradient Descent □□□□□□□□□□□□□□□□□□□□ b □□□□□□□□□□□□□□□□$g(w_t) = \frac{1}{b}\sum_{i=1}^{b} \nabla f(w_t; \xi_{t,i})$□Mini-batch Stochastic Gradient Descent □□□□

$$w_{t+1} = w_t - \frac{\eta_t}{b}\sum_{i=1}^{b} \nabla f(w_t; \xi_{t,i})$$



Mini-batch Stochastic Gradient Decent

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Unbiased Estimate

mini-batch SGD □□□□□□□□□□□□

$$\mathbb{E}_{\xi}[g(w;\xi)] = \nabla f(w)$$

Gradient Bounded Variance Assumption

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ mini-batch sample ⊠⊠⊠⊠⊠ $\nabla f(w;\xi)$□□□□□□

$$\mathrm{Var}(\nabla f(w;\xi)) \leq \sigma^2$$

Mini-batch Stochastic Gradient Decent

□□□□□□□□□□□□□□□□□□□□□□□□□□□□

$$\mathrm{Var}(g(w;\xi)) = \mathbb{E}_{\xi}[||g(w;\xi)||^2] - ||\mathbb{E}_{\xi}[g(w;\xi)]||^2 \leq \frac{\sigma^2}{b}$$

□□

$$\mathbb{E}_{\xi}[||g(w;\xi)||^2] \leq ||\nabla f(w)||^2 + \frac{\sigma^2}{b}$$

Newton's Method

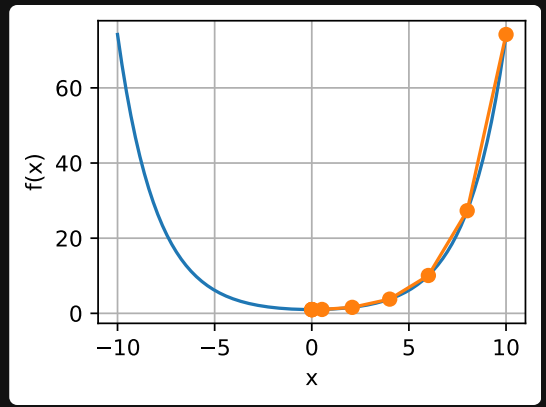
□□□□□□□□□□

$$f(x+\epsilon) = f(x) + \epsilon \nabla f(x) + \frac{1}{2}\epsilon^2 \nabla^2 f(x) + o(\epsilon^3)$$

⊠⊠ $\mathbf{H} = \nabla^2 f(x)$ ⊠⊠ Hessian ⊠⊠⊠⊠⊠⊠⊠ $n \times n$ ⊠⊠⊠⊠⊠⊠⊠ i ⊠⊠ j ⊠⊠⊠⊠⊠ $\frac{\partial^2 f}{\partial x_i \partial x_j}$□

⊠⊠⊠⊠⊠⊠ $\nabla f(x) = \frac{f(x+\epsilon)-f(x)}{\epsilon}$ ⊠ 0⊠⊠⊠ ϵ ⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠

$$\nabla f(x) + \mathbf{H}\epsilon = 0 \Rightarrow \epsilon = -\mathbf{H}^{-1}\nabla f(x)$$



Newton’s Method

⊠⊠ Hessian □□□□

Convergence Analysis

□□□□□□□□□□□□□□□ & GD□mini-batch SGD□□□□□ & mini-batch SGD□

Convergence Definition

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1. □□□□□□□□□□□□□$\mathbb{E} f(w_T) - f^* \leq \epsilon(T)$□□□ f^* □□□□□□□□□

2. □□□□□□□□□□□□$\mathbb{E}||w_T - w^*||^2 \leq \epsilon(T)$□□□ w^* □□□□□□□

□□□$\epsilon(T) \to 0$□□□□□□□□□□□□□□□□□□□

Convergence Analysis (μ-strongly convex and L-smooth & GD)

□□□□□□ f □ R^d □□μ-□□□□□□□ L-□□□□□□ $\eta \leq \frac{1}{L}$ □□□□□ w_0 □□□□ t □□□□$f(w_T)$ □ bounded □□□

$$f\left(w_t\right) - f^* \leq \left(1 - \eta\mu\right)^t \left(f\left(w_0\right) - f^*\right)$$

Proof:

□□□□□□□□

$$\begin{aligned}
f\left(w_{t+1}\right) - f\left(w_t\right) &= f\left(w_t - \eta\nabla f\left(w_t\right)\right) - f\left(w_t\right) \\
&\leq \nabla f\left(w_t\right)^\top \left(w_t - \eta\nabla f\left(w_t\right) - w_t\right) + \frac{L}{2}\left\|w_t - \eta\nabla f\left(w_t\right) - w_t\right\|^2 \\
&= -\eta\left\|\nabla f\left(w_t\right)\right\|^2 + \frac{L}{2}\eta^2\left\|\nabla f\left(w_t\right)\right\|^2 \\
&= \left(\frac{L}{2}\eta^2 - \eta\right)\left\|\nabla f\left(w_t\right)\right\|^2
\end{aligned}$$

□□□

□□□□□□□□□

$$
\begin{aligned}
f\left(w_{t+1}\right)-f^{*} &\leq(1-\eta \mu)\left(f\left(w_{t}\right)-f^{*}\right) \\
&\leq(1-\eta \mu)(1-\eta \mu)\left(f\left(w_{t-1}\right)-f^{*}\right) \\
&\leq \cdots \\
&\leq(1-\eta \mu) \cdots(1-\eta \mu)\left(f\left(w_{1}\right)-f^{*}\right) \\
&\leq(1-\eta \mu)^{t+1}\left(f\left(w_{0}\right)-f^{*}\right)
\end{aligned}
$$

⊠⊠⊠⊠⊠⊠⊠⊠ boundary⊠□□□□□□□□□

⊠⊠⊠⊠⊠⊠⊠⊠⊠⊠ $f(w_t)-f^{*} \leq \epsilon$□□□

$$
\begin{aligned}
(1-\eta \mu)^{t}(f(w_{0})-f^{*}) &\leq \epsilon \\
\Rightarrow t \log(1-\eta \mu)+\log(f(w_{0})-f^{*}) &\leq \log \epsilon \\
\Rightarrow t \log \frac{1}{1-\eta \mu}-\log(f(w_{0})-f^{*}) &\geq \log \frac{1}{\epsilon} \\
\Rightarrow t &= O(\log \frac{1}{\epsilon})
\end{aligned}
$$

⊠⊠⊠⊠⊠⊠⊠⊠⊠ GD □□□□□□□□□□

Convergence Analysis (μ-strongly convex and L-smooth & mini-batch SGD)

□□□□□□ f □ R^d □□μ-□□□□□□□ L-□□□□□□□

□□□□□□□□□

$$\mathbb{E}[f(w_{t+1}) - f(w_t)] \leq -\eta \mathbb{E}[\nabla f(w_t)^\top g(w_t;\xi_t)] + \frac{L\eta^2}{2}\mathbb{E}[||g(w_t;\xi_t)||^2] \tag{6}$$

$$\mathbb{E}[f(w_{t+1})] - f(w_t) \leq -\eta||\nabla f(w_t)||^2 + \frac{L\eta^2}{2}||\nabla f(w_t)||^2 + \frac{L\eta^2\sigma^2}{2b} \tag{7}$$

$$\leq (\eta - \frac{L\eta^2}{2})(-||\nabla f(w_t)||^2) + \frac{L\eta^2\sigma^2}{2b} \tag{8}$$

□□□ GD □□□□□□□ $\eta < \frac{1}{L}$□□□□□□□□□□

$$\mathbb{E}[f(w_{t+1})] - f(w_t) \leq \frac{\eta}{2}(-||\nabla f(w_t)||^2) + \frac{L\eta^2\sigma^2}{2b} \tag{9}$$

$$\leq \frac{\eta}{2}(-2\mu(f(w_t) - f^*)) + \frac{L\eta^2\sigma^2}{2b} \tag{10}$$

$$\leq -\eta\mu(\mathbb{E}[f(w_t)] - f^*) + \frac{L\eta^2\sigma^2}{2b} \tag{11}$$

□□□ $+f^* - f^*$□□□

$$\mathbb{E}[f(w_{t+1})] - f^* \leq (1-\eta\mu)(\mathbb{E}[f(w_t)] - f^*) + \frac{L\eta^2\sigma^2}{2b}$$

□□□□□□□□□□□□□□□□□□□□□□□□ x□□□

$$\mathbb{E}[f(w_{t+1})] - f^* + x \leq (1-\eta\mu)(\mathbb{E}[f(w_t)] - f^* + x)$$

□□□

$$(1-\eta\mu)x - x = \frac{L\eta^2\sigma^2}{2b}$$
$$x = -\frac{L\eta\sigma^2}{2\mu b}$$

□

$$\mathbb{E}[f(w_{t+1})] - f^* - \frac{L\eta\sigma^2}{2\mu b} \leq (1-\eta\mu)(\mathbb{E}[f(w_t)] - f^* - \frac{L\eta\sigma^2}{2\mu b})$$

□□

$$\mathbb{E}[f(w_t)] - f(w^*) - \frac{\eta L \sigma^2}{2\mu b} \leq (1-\eta\mu)^t \left(\mathbb{E}[f(w_0)] - f(w^*) - \frac{\eta L \sigma^2}{2\mu b} \right)$$

□□□

□□□

Convergence Analysis (non-convex & mini-batch SGD)

□□□□□□ f □ R^d □□ L-□□□□□□

□□□□□□□□□□□ t□□□

$$\frac{1}{t}\sum_{i=1}^{t}\mathbb{E}[f(w_{i+1})] - f(w_i) \leq -\frac{\eta}{2t}\sum_{i=1}^{t}||\nabla f(w_i)||^2 + \frac{L\eta^2\sigma^2}{2b} \tag{12}$$

$$\frac{1}{t}\sum_{i=1}^{t}||\nabla f(w_i)||^2 \leq -\frac{2\mathbb{E}[f(w_{t+1}) - f(w_0)]}{\eta t} + \frac{L\eta\sigma^2}{2b} \tag{13}$$

$$\leq \frac{2(f(w_0) - f(w_{\inf}))}{\eta t} + \frac{L\eta\sigma^2}{2b} \tag{14}$$

□□□

□□□ $t = O(\frac{1}{\epsilon})$□□□□□□□□□□□□□□

Other Convergence Analysis

Other Convergence Analysis

Condition	GD	SGD
Convex	$O(\frac{1}{\sqrt{T}})$	$O(\frac{1}{\sqrt{T}})$
+ Lipschitz	$O(\frac{1}{T})$	$O(\frac{1}{\sqrt{T}})$
+ Strongly Convex	$O(c^T)$	$O(\frac{1}{T})$

> 10-725/36-725: Convex Optimization(Fall 2018), Lecture 24: November 26, Ryan Tibshirani

Distributed Optimization

- Distributed Synchrounous SGD
- Distributed Asynchrounous SGD
- Federated Learning

Reference

1. 《Convex Optimization》, Stephen Boyd
2. 《Optimization Algorithm for Distributed Machine Learning》, Gauri Joshi
3. 《Dive into Deep Learning》, D2L.ai
4. 《Optimization Methods for Large-Scale Machine Learning》, Léon Bottou

Learning more about optimization and convergence analysis:

https://blog.bj-yan.top/tags/convergence-analysis/